1. Complete the following sequences:

a.	5	13	21	___	___	45
b.	26	21	16	___	___	1
c.	9	21	33	___	___	___
d.	6.5	7.2	7.9	8.6	___	___
e.	43.7	38.6	33.5	28.4	___	___

---

2. All equations balance so that the left hand side equals the same amount as the right hand side. Fill in the missing numbers to make the equations balance.

Example: 5 + 14 = 27 - ___ **Answer 8** **5 + 14 = 19**
and 27 – 8 = 19

a. 71 + 16 = 99 - ___

b. 64 ÷ 4 = 2 x ___

c. 63 – 27 = 6 x ___

d. 4 x 11 = 2 x ___

e. 154 ÷ 11 = 84 ÷ ___

---

3. Fill in the missing numbers to complete the following equivalent fractions.

a. 1/3 = __/15

b. 15/25 = 3/__

c. 9/19 = __/57

---

4. Calculate the answers to the following questions:

a. 5/6 – 3/4 = ___

b. 2/9 + 2/3 = ___

c. 7/8 + 5/24 = ___

d. 14/17 + 7/1 = ___

e. 12.09 + 1.32 = ___

f. 11.62 – 9.84 = ___

g. 9.61 x 8 = ___

h. 120.72 ÷ 4 = ___

5. Complete the following money problems.

a. A single bus fare to work costs £1.25. How much have I spent altogether on bus fares by the time I have returned home? _____

b. If I go to the newsagent and buy 5 magazines costing £3.99 each, how much did I spend in total on the magazines? _____

c. I gave the newsagent a twenty pound note for the magazines I bought in part b. How much change did I receive? _____

---

6. Answer the following calculation questions:

a. Six squared = _____

b. The cube root of 27 = _____

c. 60% of £60.00 = _____

d. 2(p – q) when p = 4.7 and q = 2.3 = _____

e. The square root of 9 = _____

f. Two fifths of 75kg = _____

g. 4(x + y) when x = 6.1 and y = 3.8 = _____

---

7. If a = 5, b = 4 and c = 2, answer the following questions:

a. c × a = _____

b. a + c – b = _____

c. a × c ÷ b = _____

d. b × a – c = _____

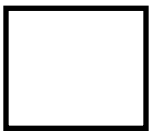
e. (b+c) × (b-c) = _____

---

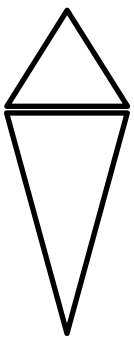
8. How many lines of symmetry do these shapes have? Some may have none.

a.

b.

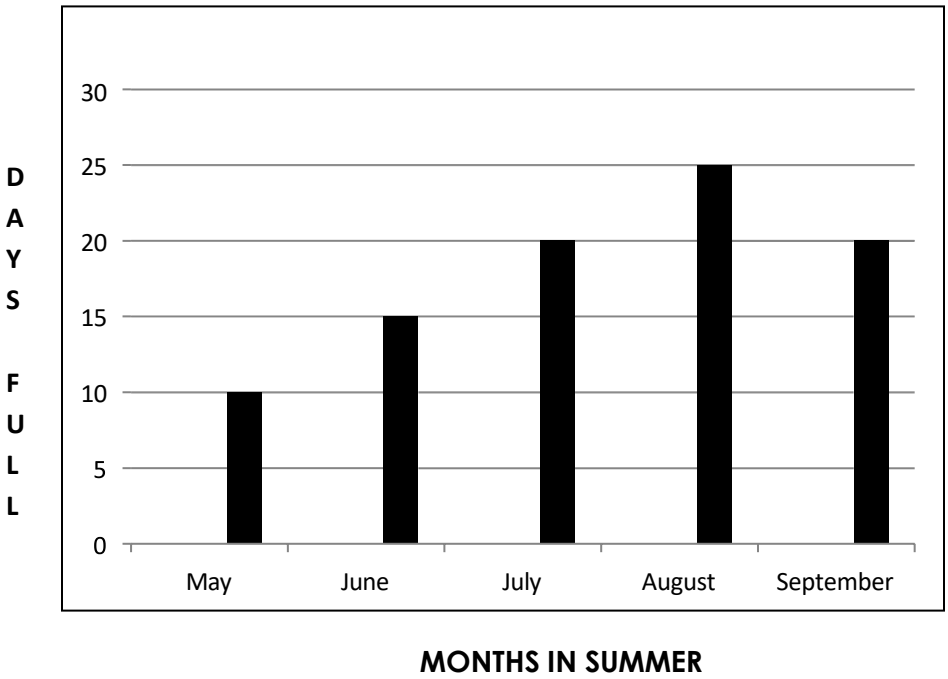


c.



d.

9. The following bar chart shows the number of days a restaurant was full during the summer in London last year.



Using the information given in the chart answer the following:

a. Which month had the most number of full days? _____

b. How many full days were there in July? _____

c. How many more full days were there in August than in May? _____

d. What was the total number of full days during the summer? _____

e. What was the average number of full days per month during the summer? _____

---

10. Underline the digit that represents the value shown.

Example: Hundreds 2 <u>5</u> 2 1

a.	Tenths	345.82	**d.**	Hundredths	256.964
b.	Units	965	**e.**	Units	658.36
c.	Tens	562.94	**f.**	Hundreds	658.23

11. Answer the following problems. Remember to use the correct units in your answers.

a. Find the total weight of 268g and 345g? _____

b. How many minutes are there in 24 hours? _____

c. How many seconds are there in a minute? _____

d. How many seconds are there in an hour? _____

e. Find the difference between 536cm than 387cm? _____

f. What is the product of 17 and 18? _____

g. Find the difference between £7.37 and £5.26? _____

h. What is the sum of £1.75, £4.36 and £2.81? _____

---

12. There are 1740 passengers waiting at Moscow Central train station. They are waiting for trains which will take them to Murmansk, St. Petersburg and Sochi.

a. If 628 are going to Sochi and 456 are going to Murmansk, how many want to go to St. Petersburg? _____

b. If each train has 700 seats including seats for the crew and each train has 12 crew, how many seats will be free on the Sochi train? _____

c. On the train to St. Petersburg 242 people are over 50, 156 are aged between 30 and 50. How many are younger than 30?

_____

d. Each second class carriage has a central aisle with three seats on either side of the aisle. There are 16 rows of seats on each side of the aisle in the passenger carriages. How many people would there be in a second class passenger carriage? _____

e. If on a train 45% of the seats are third class and for the crew and 33% are second class, how many seats would be first class? _____

13. Calculate the answer to the following squared and cubed number questions.

a. $4^3 - 2^3 =$ _____ **b.** $5^3 - 4^2 =$ _____

_____END OF TEST _____

If you have time

- **Go back and complete any questions you missed out.**
- **Check your answers carefully.**

Mathematical Reasoning questions

Read the following questions which are concerned with numbers and write your answers in the brackets.

Remember: Do these in 2 parts – stop at 'than.'

1.	15 is 3 more than half this number	( )
2.	9 is 4 more than half this number	( )
3.	18 is 4 less than twice this number	( )
4.	36 is 8 more than twice this number	( )
5.	45 is 9 less than 6 times this number	( )
6.	23 is 7 less than three times this number	( )
7.	24 is 6 less than six times this number	( )
8.	6 is 6 less than twice this number	( )
9.	14 is 10 less than half this number	( )
10.	4 is 11 less than half this number	( )

If it is smaller make it larger - multiply
If it is larger make it smaller – divide

11.	6 is 5 times smaller than this number	( )
12.	8 is 7 times smaller than this number	( )
13.	200 is 20 times larger than this number	( )
14.	If this number is made 6 times larger the answer is 42	( )
15.	8 is 20 times smaller than this number	( )
16.	14 is 7 times bigger than this number	( )
17.	4 is 9 times smaller than this number	( )
18.	250 is 10 times larger than this number	( )
19.	150 is 3 times larger than this number	( )
20.	9 is 3 times smaller than this number	( )

Now check to see how you got on!

Mean - Mode - Median - Range

Take your time to read through the examples slowly, and refer back to them when working out the problems on the next sheet.

Look at these numbers

5, 3, 6, 6, 5, 2, 1, 3, 5

- **MODE** is the number that is the MOST COMMON.
- **MEDIAN** is the MIDDLE NUMBER when all of the numbers are arranged from smallest to largest.
- **MEAN** is the TOTAL of all the numbers divided by the number of items.
- **MEAN** and ***Average*** are the same thing.
- **RANGE** is the difference between the HIGHEST AND LOWEST numbers. Don't forget that difference means you subtract.

Examples of each type are below:

- **MODE:** Look at the numbers above and you will see that 5 is the most common as there are three of them.

  Sorted. Signed_____

- **MEDIAN:** First arrange all the numbers from smallest to largest.

  1, 2, 3, 3, 5, 5, 5, 6, 6

  5 is the median (or middle) number.

  Sorted. Signed_____

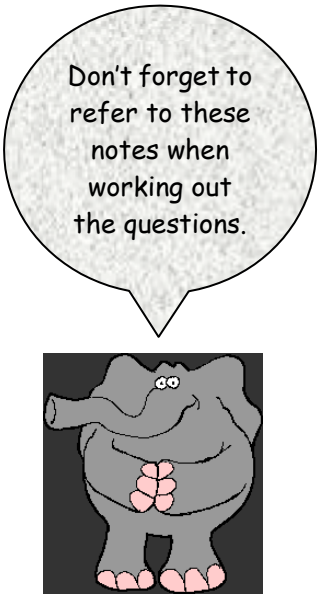
- **MEAN:** Add up all the numbers then divide by 9

  $(5+3+6+6+5+2+1+3+5) \div 9 = (36) \div 9 = 4$

  Sorted. Signed_____

- **RANGE:** $6-1 = 5$

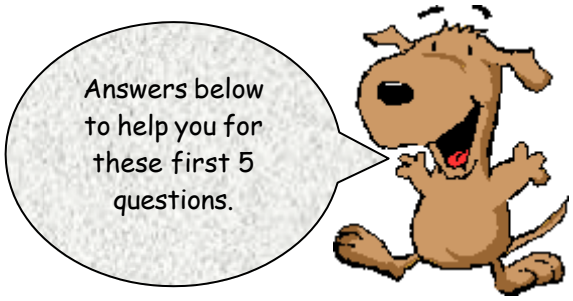
  Sorted. Signed_____



Mean - Mode - Median - Range

For each of the lists below, find the MEAN, MODE, MEDIAN and RANGE.

1. 2, 2, 5, 7, 9
2. 4, 5, 8, 8, 10, 11, 17
3. 10, 13, 13, 16, 16, 16, 21
4. 25, 27, 28, 29, 29, 29, 32, 35, 36
5. 37, 38, 35, 42, 35, 37, 35



Answers: 1.	Answers: 2.	Answers: 3.	Answers: 4.	Answers: 5.
Mean – 5	Mean – 9	Mean – 15	Mean – 30	Mean – 37
Mode – 2	Mode – 8	Mode – 16	Mode – 29	Mode – 35
Median – 5	Median – 8	Median – 16	Median – 29	Median – 37
Range - 7	Range - 13	Range - 11	Range - 11	Range - 5

Worded Problems

1. The label on a box of matches says there are, on average, 25 matches in each box. In a DIY store the shopkeeper checks a few boxes to see if this is true. Listed below is the number of matches he found in the boxes he checked: 25, 20, 35, 27, 26, 24, 32, 26, 28. **Work out the** *mean number* **of matches in the boxes that he checked.** **Answer:** _____

2. My mum went shopping and bought lots of potatoes. I took some out and weighed them. What I found is listed below:

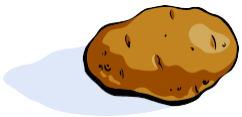
   100g, 85g, 95g, 100g, 96g

   A. Work out the median weight Answer: _____

   B. Work out the mean weight Answer: _____

   (B has a decimal answer)

   C. What is the range? Answer: _____



Sorted! Signed _____

Answers for these last two are in the answer sheets. No peeping please.

CORE SKILLS 6

1

A full circle = 9 and a half circle = half of 9. Choose the number for Thursday from the options provided.

Monday	
Tuesday	
Wednesday	
Thursday	
Friday	

27 22.5 24.5 20 24

2

Haley left for a walk at 11:30 am. She walked for 1 hour 20 mins then stopped for a cup of coffee for 30 minutes. She then continued walking for 45 mins. What time did she finish walking?

3

A map of France has a scale of 1:300,000. What is the real distance represented by 2cm on the map?

4

Below is a number rule.

Add 0.5 to the previous number and multiply by 2.

Jim uses this rule to write a sequence of six numbers. The first number is 6. The third number is 27. What is the second number in the sequence?

5

What is the fourth number in the sequence from Question 4?

6

What is the sixth number in the sequence from Question 4?

7

450 people went to the theatre to see a performance of Macbeth. 70% of them had been to the theatre before. How many had never been to the theatre before?

CORE SKILLS 6

8	10 boxes need to be stored on top of each other. The first box is 70 cm high and each subsequent box is 5 cm less in height than the previous one. What is the total height (in metres) of all 10 boxes when they are stacked on top of each other?
9	What is the value of Y in the equation: $8YY + 7 = 135$
10	John buys some new printer ink for his printer as he needs to print three copies of his 500 page book that contains 60 coloured pages. If a coloured ink cartridge prints 200 pages before it needs replacing and a black ink cartridge prints 1000 pages before it needs replacing, how many cartridges does John need to purchase?

ANSWERS - COMPLETE MATHS – PART 26

Maths

Maths type 1 paper 1

Page 1

1.

a. 29, 37
b. 11, 6
c. 45, 57, 69
d. 9.3, 10
e. 23.3, 18.2

2.

a. 12
b. 8
c. 6
d. 22
e. 6

3.

a. 5
b. 5
c. 27

4.

a. $^{1}/_{12}$
b. 8/9
c. 1 1/12
d. 7 $^{14}/_{17}$
e. 13.41
f. 1.78
g. 76.88
h. 30.18

Page 2

1.

a. £2.50
b. £19.95
c. 5p

2.

a. 36
b. 3
c. £36
d. 4.8
e. 3
f. 30kg
g. 39.6

3.

a. 10
b. 3
c. 2.5
d. 18
e. 12

4.

a. 2
b. 4
c. 1
d. none

Page 3

5.

a. August
b. 20
c. 15
d. 90
e. 18

6.

a. 345.82
b. 965
c. 562.94
d. 256.964
e. 658.36
f. 658.23

Page 4

7.

a. 613g
b. 1440
c. 60
d. 3600
e. 149cm
f. 306
g. £2.11
h. £8.92

8.

a. 656
b. 60
c. 258
d. 96
e. 22%

Page 5

9.

a. 56
b. 109

Maths Reasoning sheet

1.	15 is 3 more than / **half this number** **12 is**	24
2.	9 is 4 more than / **half this number** **5 is**	10
3.	18 is 4 less than / **twice this number** **22 is**	11
4.	36 is 8 more than / **twice this number** **28**	14
5.	45 is 9 less than/ **six times this number** **54**	9
6.	23 is 7 less than / **three times this number** **30**	10
7.	24 is 6 less than / **six times this number** **30**	5
8.	6 is 6 less than / **twice this number** **12**	6
9.	14 is 10 less than / **half this number** **24**	48
10.	4 is 11 less than / **half this number** **15**	30

11	6 is 5 times smaller than/ this number	30
12.	8 is 7 times smaller than/ this number	56
13.	200 is 20 times larger than/ this number	10
14.	If this number is made 6 times larger the answer is 42	7
15.	8 is 20 times smaller than/ this number	160
16.	14 is 7 times bigger than/ this number	2
17.	4 is 9 times smaller than/ this number	36
18.	250 is 10 times larger than/ this number	25
19.	150 is 3 times larger than/ this number	50
20.	9 is 3 times smaller than/ this number	27

Mean, Median, Mode and Range

(Answers to questions 1-5 are given on the question sheet)

1. 27
2. A. 96g
   B. 95.2g
   C. 15g

Core Skills 6

1. 22.5
1. 2:05 pm
2. 600,000 cm or 6 km
3. second number is 13
4. fourth number is 55
5. sixth number is 223
6. 135
7. 4.75m
8. 16
9. John needs to buy 1 coloured ink cartridge and 2 black ink cartridges.